

Problem Set 3 – Solution

Microeconomic Theory 3 – ECON 709, Fall 2026

This document provides a solution sketch for each answer. Sometimes, the suggested solution is of course not the only "right" answer, but you might have come up with different, equally correct answers.

The solutions for the Rubinstein problems are taken from Rubinstein's solution manual.

1. (Rubinstein, Lecture 2, Problem 1.a-c)

- (a) Is the following statement correct: "if both U and V are utility representations of $\succsim$, then there is a strictly monotonic function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $V(x) = f(U(x))$ "?

Solution:

The statement is not correct:

Let $X = \mathbb{R}$ and preferences be represented by the utility functions

$$V(x) = x \text{ and } U(x) = \begin{cases} x + 1 & \text{if } x > 0 \\ x & \text{if } x \leq 0. \end{cases}$$

The only increasing function $f : \mathbb{R} \rightarrow \mathbb{R}$ that satisfies $V(x) = f(U(x))$ is

$$f(x) = \begin{cases} x & \text{if } x \leq 0 \\ 0 & \text{if } 0 < x \leq 1 \\ x - 1 & \text{if } 1 < x \end{cases}$$

which is not strictly increasing. Note that for V also to be a utility representation, f only has to be strictly increasing on the set of values of U .

- (b) Can a continuous rational preference relation be represented by a discontinuous utility function?

Solution:

Yes. Example: Let $X = \mathbb{R}$ and consider $\succsim$ such that $x \succsim y$ if $x \geq y$. The preferences are continuous: Take any a, b such that $a \succ b$ and let $\epsilon \in (0, 1/2(a - b))$. Then $x \succ y$ for all $x \in \text{Ball}(a, \epsilon)$ and all $y \in \text{Ball}(b, \epsilon)$. The following discontinuous function represents $\succsim$:

$$u(x) = \begin{cases} x + 1 & \text{if } x > 0 \\ x & \text{if } x \leq 0 \end{cases}$$

- (c) Show that in the case of $X = \mathbb{R}$, the rational preference relation that is represented by the discontinuous utility function $u(x) = \lfloor x \rfloor$ (the largest integer n such that $x \geq n$) is not a continuous relation.

Solution:

It is probably easiest to show that C1 is violated. For practice let's look at C2. Consider sequences $(.9 - .4/n)_{n=1, \dots, \infty}$ and $(1 - .4/n)_{n=1, \dots, \infty}$. We have that $.9 - .4/n \succsim 1 - .4/n$ for all n and $(.9 - .4/n) \rightarrow .9$ and $(1 - .4/n) \rightarrow 1$. But $1 \succ .9$, and not $.9 \succsim 1$. Thus, C2 is violated.

2. Let $\succsim$ be a continuous and rational preference relation on X , which is a convex subset of $\mathbb{R}^n$. Assume that $\succsim$ is not the total indifference. Show that if $x \succ y$, then there exists $z \in X$ on the interval between x and y such that $x \succ z \succ y$.

Hint: You can use a proof by contradiction which makes use of a strategy related to the one in the first step in our proof of Debreu's theorem.

Solution:

The proof can be found on p. 18 of Rubinstein's textbook. Let I be the interval between x and y . Assume by contradiction that there is no $z \in X$ in I such that $x \succ z \succ y$. Then $z \succsim x$ or $y \succsim z$ for all $z \in I$.

Construct inductively two sequences of points in I , $\{x_t\}$ and $\{y_t\}$, in the following manner: First, define $x_0 = x$ and $y_0 = y$. Assume that the two points x_t and y_t are defined, belong to I , and satisfy $x_t \succsim x$ and $y \succsim y_t$. Consider m , the middle point between x_t and y_t . Either $m \succsim x$ or $y \succsim m$. In the former case, define $x_{t+1} = m$ and $y_{t+1} = y_t$, and in the latter case define $x_{t+1} = x_t$ and $y_{t+1} = m$. The sequences $\{x_t\}$ and $\{y_t\}$ are converging, and they must converge to the same point z because the distance between x_t and y_t converges to zero. By the continuity of $\succsim$, we have $z \succsim x$ and $y \succsim z$ and thus, by transitivity, $y \succsim x$, which contradicts the assumption that $x \succ y$.

3. Let $X = \mathbb{N} \times \mathbb{R}$, that is x_1 is a natural number (this can be interpreted as element 1 being an indivisible good) and x_2 is a real number. Suppose the preference relation over X is the lexicographic preference $\succsim_L$ defined by: $(x_1, x_2) \succsim_L (y_1, y_2)$ iff $x_1 > y_1$ or both $x_1 = y_1$ and $x_2 \geq y_2$.

- (a) Is $\succsim_L$ continuous? Does $\succsim_L$ have a utility representation? If yes, give an example.

Solution:

We show that $\succsim_L$ satisfies $C1$. Take any $a, b \in X$ such that $a \succ b$. Then either 1) $a_1 > b_1$ or 2) $a_1 = b_1$ and $a_2 > b_2$. If 1), then $x \succ y$ for all $x \in \text{Ball}(a, r)$ and $y \in \text{Ball}(b, r)$ where $r = 1/3(a_1 - b_1)$. If 2), then $x \succ y$ for all $x \in \text{Ball}(a, r)$ and $y \in \text{Ball}(b, r)$ where $r = 1/3(|a_2| - |b_2|)$.

A utility function which represents $\succsim_L$ is

$$u(x) = \begin{cases} x_1 + 1/3 \frac{x_2}{x_2+1} & \text{if } x_2 \geq 0 \\ x_1 - 1/3 \frac{|x_2|}{|x_2|+1} & \text{if } x_2 < 0 \end{cases}$$

Let $a \succ b$. If 1), then $u(a) > a_1 - 1/3 > b_1 + 1/3 > u(b)$ as $a_1 \geq b_1 + 1$. If 2), then $u(a) > u(b)$ because the first term in the

utility function is the same for a and b and the second term is strictly larger for a . Let $a \sim b$, then $a = b$ and $u(a) = u(b)$.

- (b) Now suppose that $X = \mathbb{R} \times \mathbb{N}$, and the preference relation is given by $\succsim_L$ as above. Is $\succsim_L$ continuous? Does $\succsim_L$ have a utility representation? If yes, give an example.

Solution:

The preferences are not continuous:

Consider $(1, 2) \succ (1, 1)$. However, $(1, 1) \succ (1 - \epsilon, 2)$ for any $\epsilon > 0$. Thus, C1 cannot be satisfied.

There does not exist a utility representation of $\succsim_L$. You can follow the proof from the lecture where we showed that lexicographic preferences on the unit square do not have a utility representation.